v.
4/9/2026

PetDx®

The Liquid Biopsy Company for Pets ™

PetDx

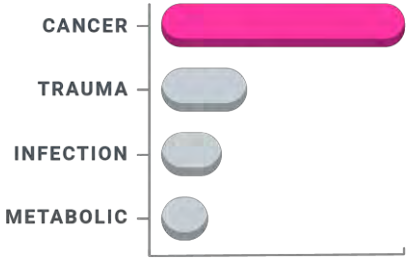
STIFEL

Unleashing the Power of Genomics to Improve Pet Health™

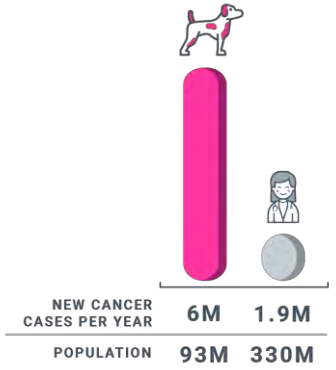
For an expanded list of references, visit petdx.science/oncok9-veterinary-overview and petdx.com/resources

Cancer is the Single Largest Unmet Need in Veterinary Medicine

Cancer is by far the **leading cause of death in dogs**



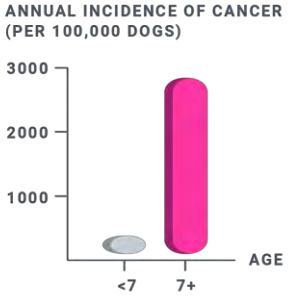
Dogs have a **tenfold higher annual incidence** of cancer relative to humans



Dogs have a **1 in 3 Lifetime Risk** of Cancer

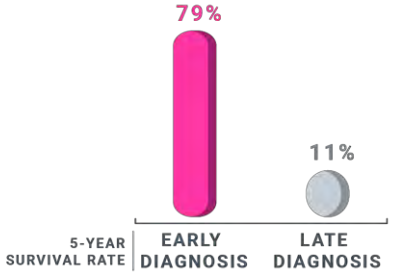
ALL DOGS

DOGS >10 YRS OLD



9x Higher Risk Of Cancer Over Age 7

Early detection methods have **drastically improved outcomes** for humans



Sources: seer.cancer.gov, J Vet Intern Med. doi: 10.1111/j.1939-1676.2011.0695.x. Vet J. doi: 10.1016/j.tvjl.2016.03.009. American Veterinarian, National Canine Cancer Foundation, American Cancer Society. AAHA: https://www.aaha.org/your-pet/pet-owner-education/ask-aaha/canine-cancer/ AKC-CHF: https://www.akcchf.org/canine-health/top-health-concerns/canine-cancer/cancer-in-the-dog.html ACF: https://acfoundation.org/the-10-warning-signs-of-cancer/ AVMA: https://www.avma.org/resources/pet-owners/petcare/cancer-pets

PetDx

There is no established cancer screening paradigm for dogs. Current methods are inadequate for early detection of cancer.

Early detection brings care back to >100,000 GPs *vs* ~450 Boarded Vet Oncologists

> *... early detection is critical for the best outcome ... half of all canine cancers are treatable if caught early enough.*

AAHA

> *... early detection and treatment are the best ways to manage neoplasia in pets... neoplasia is frequently treatable...*

AVMA American Veterinary Medical Association

Sources: seer.cancer.gov, J Vet Intern Med. doi: 10.1111/j.1939-1676.2011.0695.x. Vet J. doi: 10.1016/j.tvjl.2016.03.009. American Veterinarian, National Canine Cancer Foundation, American Cancer Society. AAHA: https://www.aaha.org/your-pet/pet-owner-education/ask-aaha/canine-cancer/ AKC-CHF: https://www.akcchf.org/canine-health/top-health-concerns/canine-cancer/cancer-in-the-dog.html ACF: https://acfoundation.org/the-10-warning-signs-of-cancer/ AVMA: https://www.avma.org/resources/pet-owners/petcare/cancer-pets

PetDx is Rapidly Unlocking the Oncology Opportunity

Unmatched industry expertise and capabilities

illumina
zoetis
ANTECH
labcorp
Memorial Sloan Kettering Cancer Center
MARS Petcare
IDEXX

Validated technology, clinical utility + industry reputation + in-house lab and bioinformatics platform

Ongoing

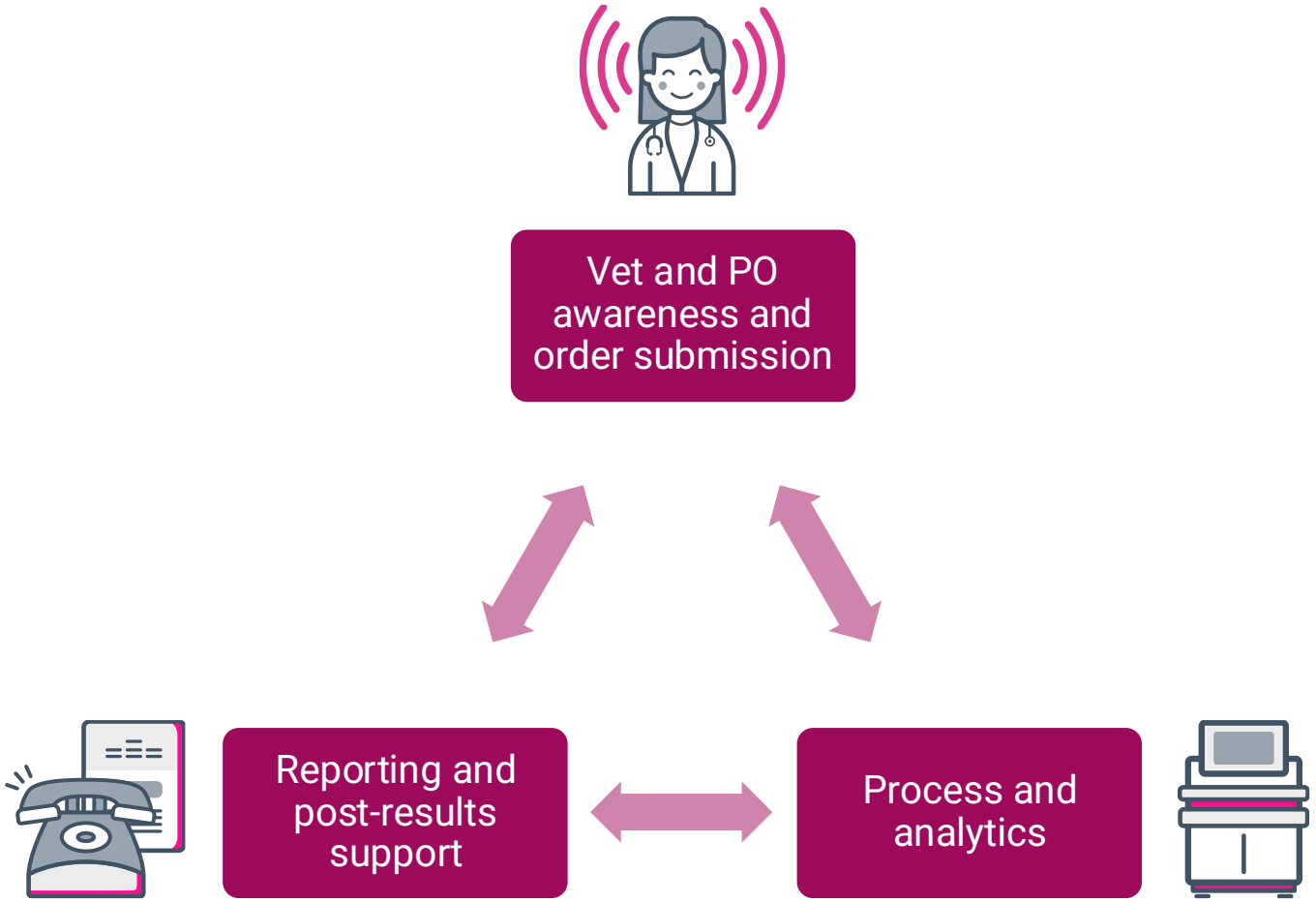
Commercial validation and scaling capabilities

- Early Go-to-market Success
- Leading Industry Partnerships
- State-of-the-Art NGS Lab

Sources: Company data.

PetDx

Our Modern Technology Platform is Purpose Built to Service Veterinarians



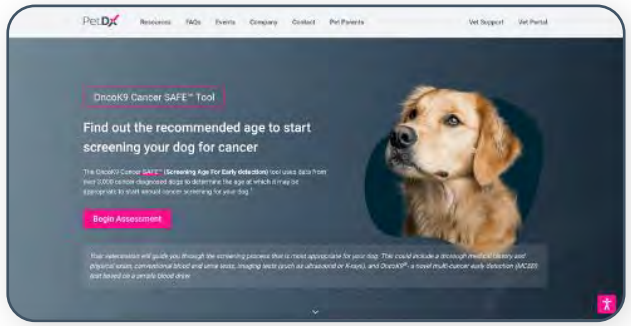
Our Modern Technology Platform is Purpose Built to Service Veterinarians

Vet and PO awareness and order submission

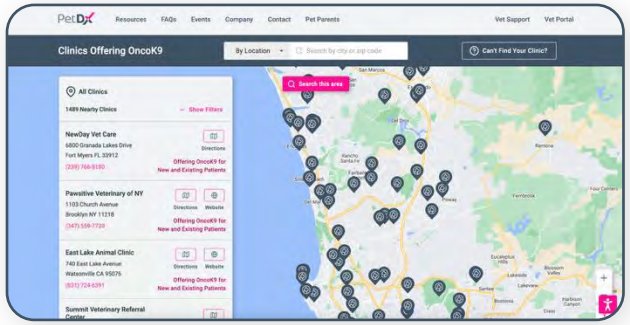
Process and analytics

Reporting and post-results support

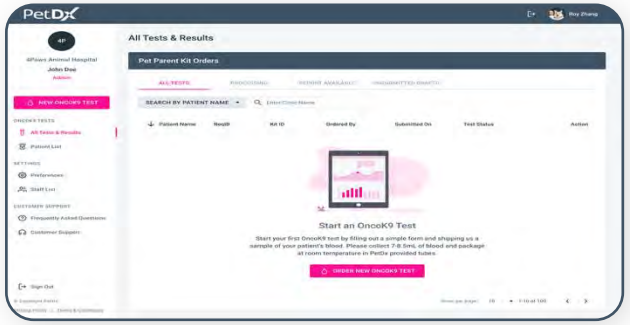
OncoK9 Cancer SAFE™ Tool



Clinic Locator



Vet Portal



Sources: Based on PetDx proprietary software stack developed or in-development as of Q4 '22.
* PAD/DRP: internally developed Pipeline Automation Dashboard and Data Review Platform
** LIMS: internally developed Laboratory Information Management System

Our Modern Technology Platform is Purpose Built to Service Veterinarians

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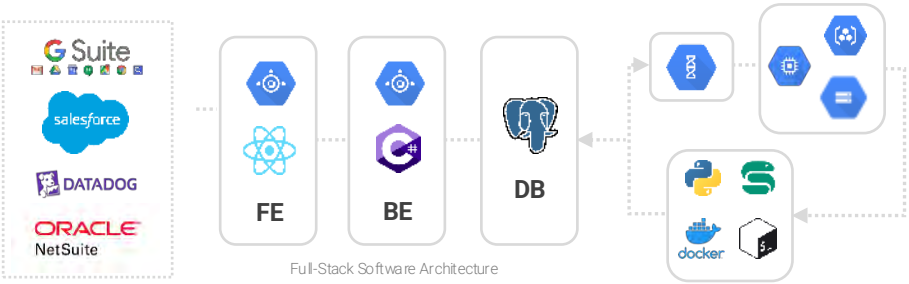
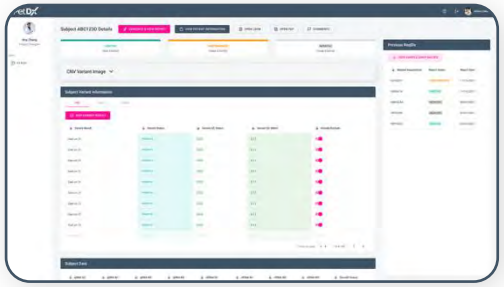
Modern NGS Development Architecture

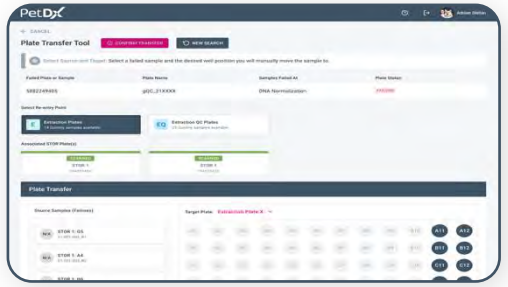


Powered by: Google Cloud

PAD/DRP*



LIMS/Automation Platform**



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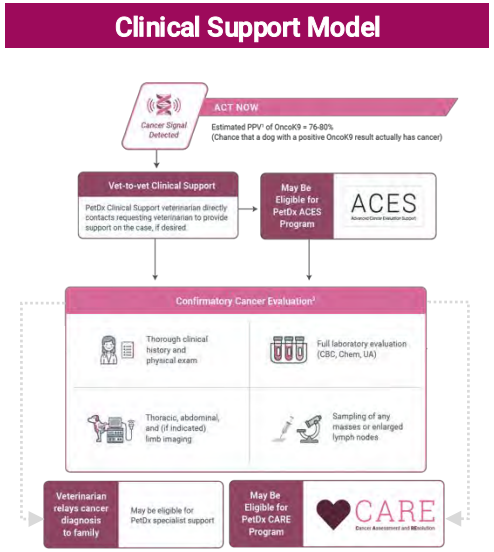

Streamlined/Integrated Electronic Report Delivery

- Easy-to-interpret results
- End-to-end database integrated
- Delivered electronically via Vet Portal or PMS
- Provides clinical workup suggestions
- End-User, Admin, Vet, Vet Clinic interfaces

Sources: Based on PetDx proprietary software stack developed or in-development as of Q4 '22.
* PAD/DRP: internally developed Pipeline Automation Dashboard and Data Review Platform
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PetDx

We Continue Driving Domain Leadership and Building a Reputation for Scientific Excellence

Clinical Study: CANcer Detection in Dogs (CANDiD)

Largest peer-reviewed clinical validation study ever published in veterinary oncology diagnostics.

1000+
client-owned dogs

85
breeds

40+
cancer types

40+
study sites

77
author block

- 39 DVMs, 3MDs
- 21 boarded specialists
- 14 PhDs

Study Collaborators:

FLINT ANIMAL CANCER CENTER COLORADO STATE UNIVERSITY

vca animal hospitals

UNIVERSITY of GUELPH ONTARIO VETERINARY COLLEGE

THE OHIO STATE UNIVERSITY COLLEGE OF VETERINARY MEDICINE

COLLEGE OF VETERINARY MEDICINE

College of Veterinary Medicine University of Missouri

PetDx

Program	Design	Enrollment	Analysis	Next Milestone
Scientific Review Vet onc. & genomics review publication *Review*	✓ Published Mar '21			
Proof of Concept POC publication *Screening, AiD*	✓ Published Jul '21			
CANDiD Study Validation Study *Screening, AiD*	✓ Published Apr '22			
Age at Cancer Dx Cancer screening age study, SAFE tool *Screening*	✓ Published Feb '23			
Clinical Experience Clinical experience and outcomes study *Observed Outcomes*	✓ Published Mar '23			
Post-Dx Use Cases Demonstrate performance in additional use cases *MRD / RM / Tx Response*	✓ Completed H1 '23			*Paper submission*
CLASSIC / CLASSIC+ Prospective Lifetime Screening Study *Screening, Cumulative Detection, Outcomes*	*Indefinite*			*Target Enrollment: 200+ patients*
LIQUiD LIQuid biopsy Utility in Dogs *AiD*				*Target Enrollment: 200 patients*
COSMiC Common Surgically Managed cancers in Canines *Screening, AiD, MRD*				*Target Enrollment: 100 patients*
ADaPT Assay Development and Process Testing *Assay dev / refinement*				*Site Recruitment: ~10 clinics*
OncoFeline™ Feline Liquid Biopsy Clinical Validation Study *Screening, AiD*				*Study design & IACUC approval*

Canine Studies

Feline Study

In Q1 ‘23, Vet Industry Leaders Join to Steer Commercial Roadmap

Animal Health sales, marketing and operations

Alejandro Bernal, DVM, MBA
Chief Executive Officer,
Board Member
zoetis MARS Petcare

Paul Bergkamp
Chief Commercial Officer
IDEXX covetrus

Lab operations and veterinary reputation

Andi B. Flory, DVM, DACVIM (Onc)
Chief Medical Officer
ETHOS VETERINARY HEALTH · Cornell University Hospital for Animals · AMC Schwarzman Animal Medical Center

Susan Hicks, CGMBS, MB (ASCP)
VP, Laboratory Operations
HOLOGIC labcorp sequenom

Science, bioinformatics and start-up foundations

Daniel S. Grosu, MD, MBA
Founder, Sr. Advisor, BoD
Chairman, ex-CMO Illumina
illumina sequenom Johnson & Johnson · University of Oxford

Dana WY Tsui, PhD
Chief Scientific Officer
CUHK · Cancer Research UK · Weill Cornell Medicine · Memorial Sloan Kettering Cancer Center

Ilya Chorny, PhD
Chief Technology Officer
illumina UCSF University of California San Francisco

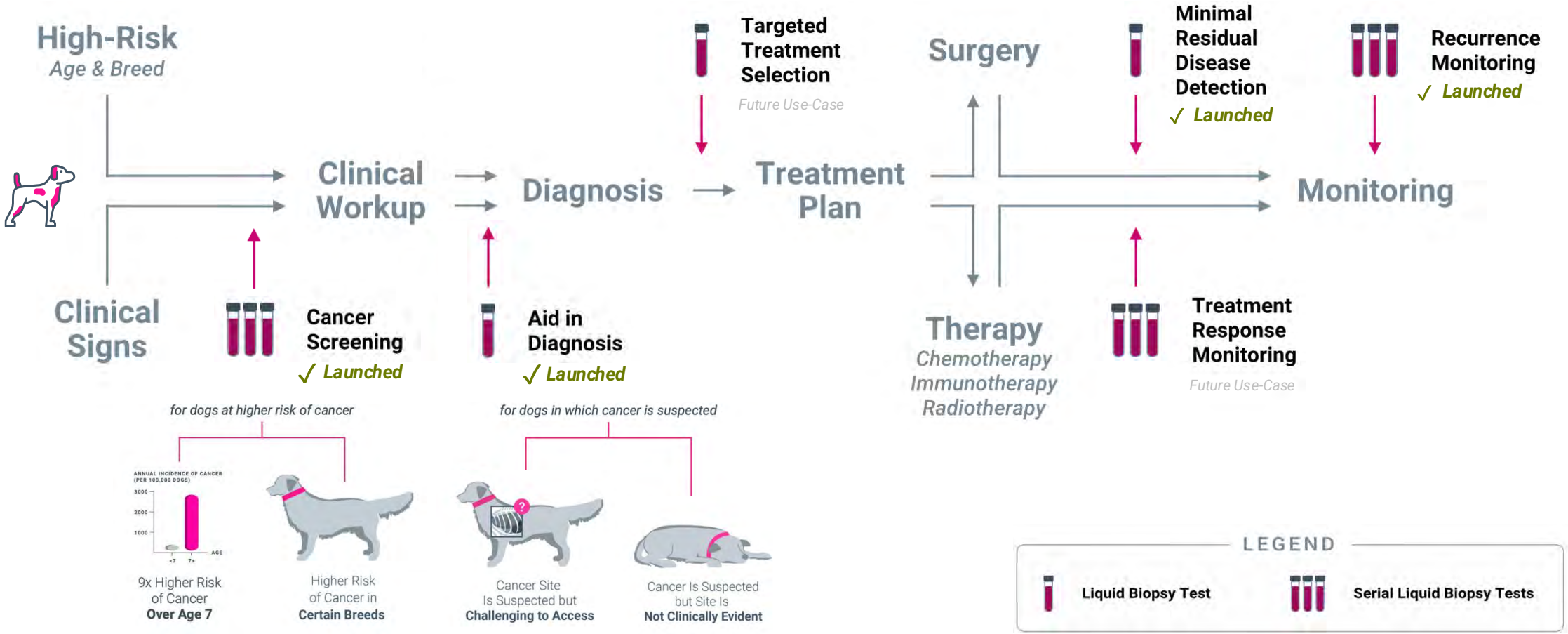
Kalle Marsal, MS, MBA
Chief Operating Officer,
Board Member
BAIN & COMPANY · hp · Mitek · Stanford

Arthur Polk
Chief Financial Officer, Director,
Corporate Development
SIEMENS · Wisconsin School of Business, University of Wisconsin–Madison

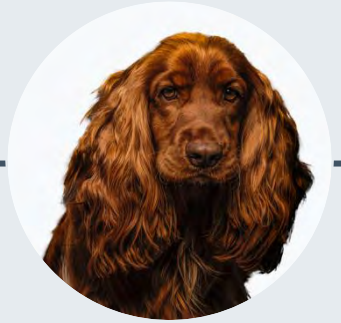
PetDx

OncoK9 is now Clinically Validated in Multiple Key Use-Cases

Each use case offers new, material revenue potential to vets



PetDx



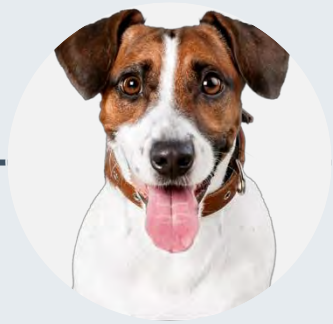
Molly

- 15-Year-Old
- Female Spayed
- Cocker Spaniel
- 16 lbs

Case Summary

- Owner wanted to screen for cancer because of recent loss of littermate
- OncoK9: ***Cancer Signal Detected***
- Exam, labs, and chest x-rays were normal
- Abdominal ultrasound = enlarged hepatic lymph nodes and changes in liver, needle aspirate performed on lymph nodes
- Cytology of hepatic lymph nodes = lymphoma
- This case of hepatic lymphoma was diagnosed before clinical signs, whereas this cancer is typically diagnosed when patients are very sick and in organ failure
- Molly was able to start chemotherapy while she was still feeling well

PetDx



Whiskey

- 12-Year-Old
- Female Spayed
- Jack Russell Terrier
- 36 lbs

Case Summary

- Patient had a screening ultrasound and masses in the spleen and liver were found – surgery to remove the spleen and to perform a small liver biopsy were performed
  - Histopathology confirmed a benign splenic lesion but was unable to determine if the liver mass was benign or malignant
- OncoK9 was used after surgery to evaluate for whether this was a malignancy
- OncoK9: ***Cancer Signal Detected***
  - The OncoK9 result helped the veterinarian and family understand the clinical situation
- A second surgery was elected and the liver mass was removed
  - Histopathology was consistent with completely excised hepatocellular carcinoma, a cancer that carries an excellent prognosis
- Whiskey is doing great and PetDx was ecstatic to be able to help

PetDx

Pet Oncology Category is Showing Strong Proof of Market Readiness

Animal Health markets historically exhibit **Clinical Bandwagons**, resulting from the intersection of prevention, convenience, innovation and economics.

Clinical Bandwagons Drove Multi-Billion Dollar Markets

VANGUARD by zoetis	Vaccination (prevention)
NexGard FLEA & TICK PROTECTION NEVER TASTED SO GOOD; Simparica (sarolaner) Chewables	Tick-borne diseases (testing and prevention)
SNAP®	In-house equipment and rapid tests
apoquel (oclacitinib tablet)	Atopic dermatitis (itching; comfort)
Librela® Bedinvetmab	Monoclonal antibodies (pain)

PetDx

Oncology Dx / Tx is Exhibiting Inflection & Momentum

Cancer is Single Largest Unmet Need in Veterinary Medicine

2020 - 2023
Multiple First Clinical Cancer-Specific Product Announcements

Leading Scientific and Market Research Validate Opportunity

PetDx
5 Key Industry Scientific Publications

Brakke CONSULTING
2022 Cancer in Dogs and Cats

VCA Animal Hospitals
2022 VCA Pet Cancer Alliance Gallup Study

Nationwide
About the Size of It: Scaling Canine Cancer Risk

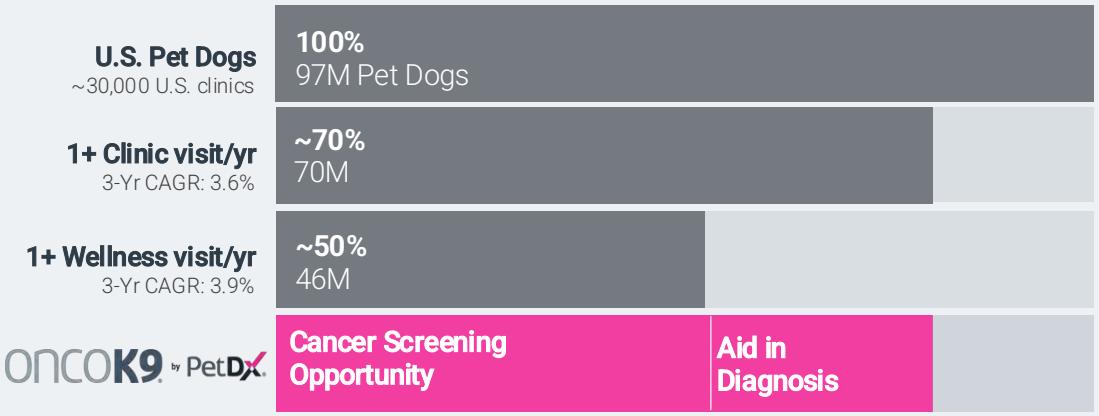
Stanford AIMI · One Health Company
2023: Nature – Analyses of canine cancer mutations and treatment outcomes using real-world clinico-genomics data of 2119 dogs

PetDx is Leading the Next Clinical Bandwagon of the Animal Health Industry: Cancer

PetDx is a First-Mover in this Tailwind Category

Multi-Billion Dollar Market Opportunity

Illustrative Market Sizing - US Pet Dogs TAM



Est. 20 - 30M U.S. Pet Dogs in high-risk breed or age intended use population for Annual Cancer Screening

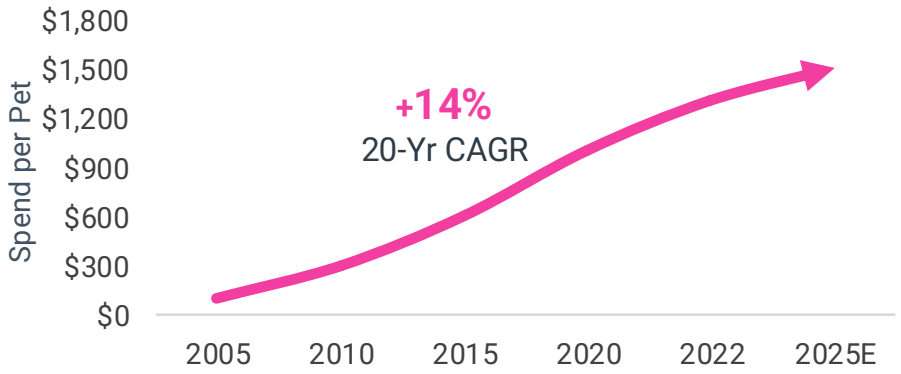
Est. >12M U.S. Pet Dogs eligible for Aid in Diagnosis

Multiple Disruptive, Secular Market Growth Drivers

- Humanization of Pets
- Emergence of Pet Generation
- Growing Adoption & Lifespan
- Consolidation of Vet Clinics
- Growth in Wellness Visits & Preventive Dx
- Lack of Regulation for non-POC Dx – No FDA, USDA

Rising Spend per Pet

Household Annual Spend per Pet (U.S. Average)



Sources: Morgan Stanley Research, Harris Poll Survey, IDEXX Practice Intelligence Data, VCA, AAHA, APPA, Brakke Consulting, AVMA Sourcebook, Packaged Facts, NAVC, and JAVMA DOI:10.2460/javma.247.1.40, BMC DOI: 10.1186/s40575-020-00086-8. Screening and AiD data through 2/14/23. Spend per Pet reflects BLS, APPA, and Morgan Stanley Data.

PetDx

First Year Go-to-Market has Built Momentum

AMC Schwarzman Animal Medical Center SINCE 1910

The world's largest nonprofit animal hospital with a leading veterinary oncology program driving patient care and research.

bluepearl

Part of Mars Petcare, BluePearl is a leading U.S. operator of emergency and specialty veterinary hospitals across 28 states.

FLINT ANIMAL CANCER CENTER COLORADO STATE UNIVERSITY

The Flint Animal Cancer Center is a world leader in pet cancer treatment and comparative oncology research.

vca animal hospitals

Part of Mars Petcare, VCA manages 1,000+ hospitals in N. America with a leading clinical partnership program.

petco vetco total care THE HEALTH + WELLNESS CO.

Omni-channel pet health providers with 150+ veterinary hospitals providing care for pets across the US.

MVP, Pet Vet Care, Blue River Pet Care, VetCor, pathway, People, Pets & Vets, AMERIVET, Lakefield, encore VET GROUP

Direct sales relationships with more than 700 independent or consolidator-owned vet clinics in 47 states, Puerto Rico and 3 Canadian provinces.

ANTECH BETTER DIAGNOSTICS. BETTER MEDICINE.™

Servicing: vca animal hospitals, bluepearl, Banfield PET HOSPITAL

Part of Mars Petcare, Antech Diagnostics is a leading veterinary distributor serving more than 17,000 customers including VCA and Banfield.

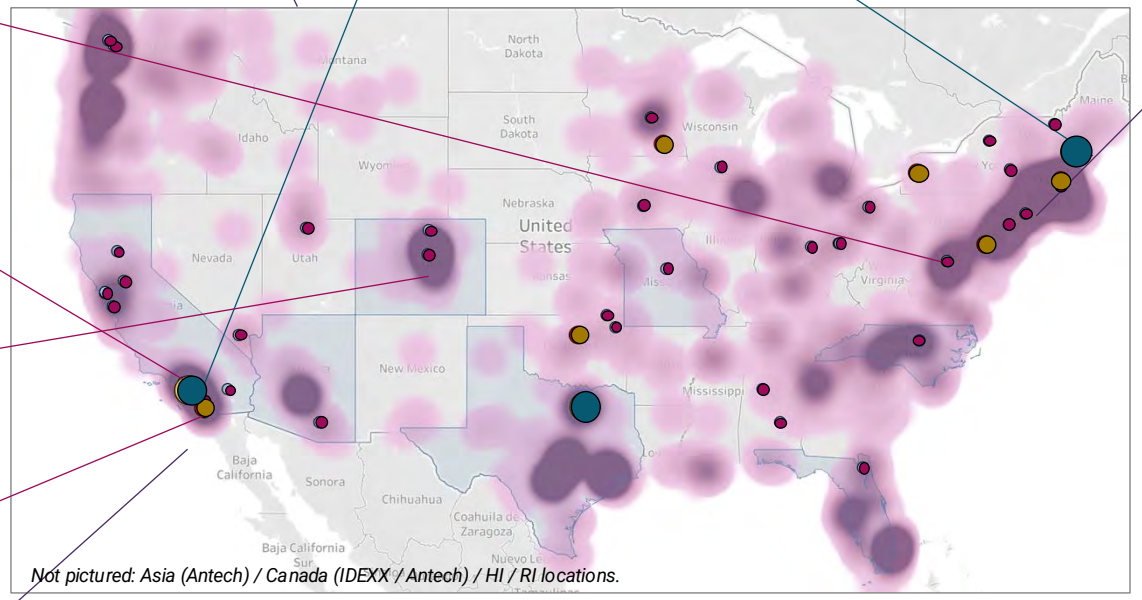
IDEXX LABORATORIES

Servicing: NVA

The largest veterinary diagnostics distributor globally, IDEXX services over 20,000 clinics in North America including NVA.

small door VETERINARY

Modern, membership-based veterinary care company with a mission to be a socially responsible employer and pet-care provider.



Not pictured: Asia (Antech) / Canada (IDEXX / Antech) / HI / RI locations.

>25,000 clinics reached, driving **>1,200** directly contracted practices (**>5,000** veterinarians and **>45** consolidators represented)

SELECT CUSTOMERS AND PARTNERS BY SEGMENT

Clinical Study Sites | Research Partners | Direct Clinics | Distributors

Dogs Sequenced

Clinical **5,000+**

R&D **2,500+**

Consumers Engaged

- Pet Parent Marketing
- Cancer SAFE tool
- Clinic Locator

Four Demand Vectors

Outbound Inbound

Consolidator Distribution

U.S. Clinics Reached **>25,000**

Veterinarians Engaged

Leads Generated **500k+**

Test Results Issued **40k+**

Vet Earned Media **300+**

KOL mentions **50+**

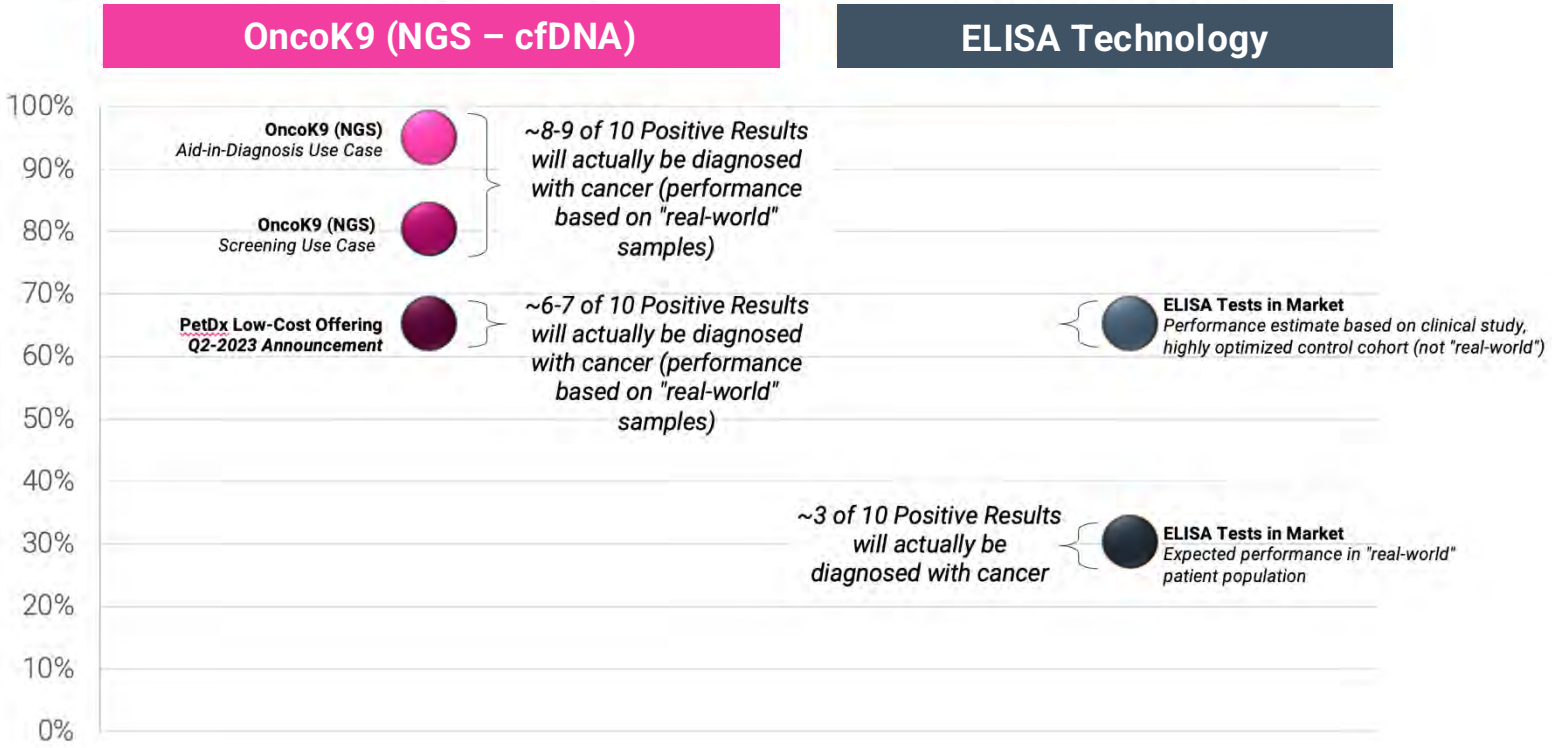
PetDx

PetDx clinic and test database as of 02/2023

OncoK9 NGS Technology Comparison to ELISA methods

Probability that a patient with a Positive Result has cancer (PPV):

False positives: A wild goose chase!



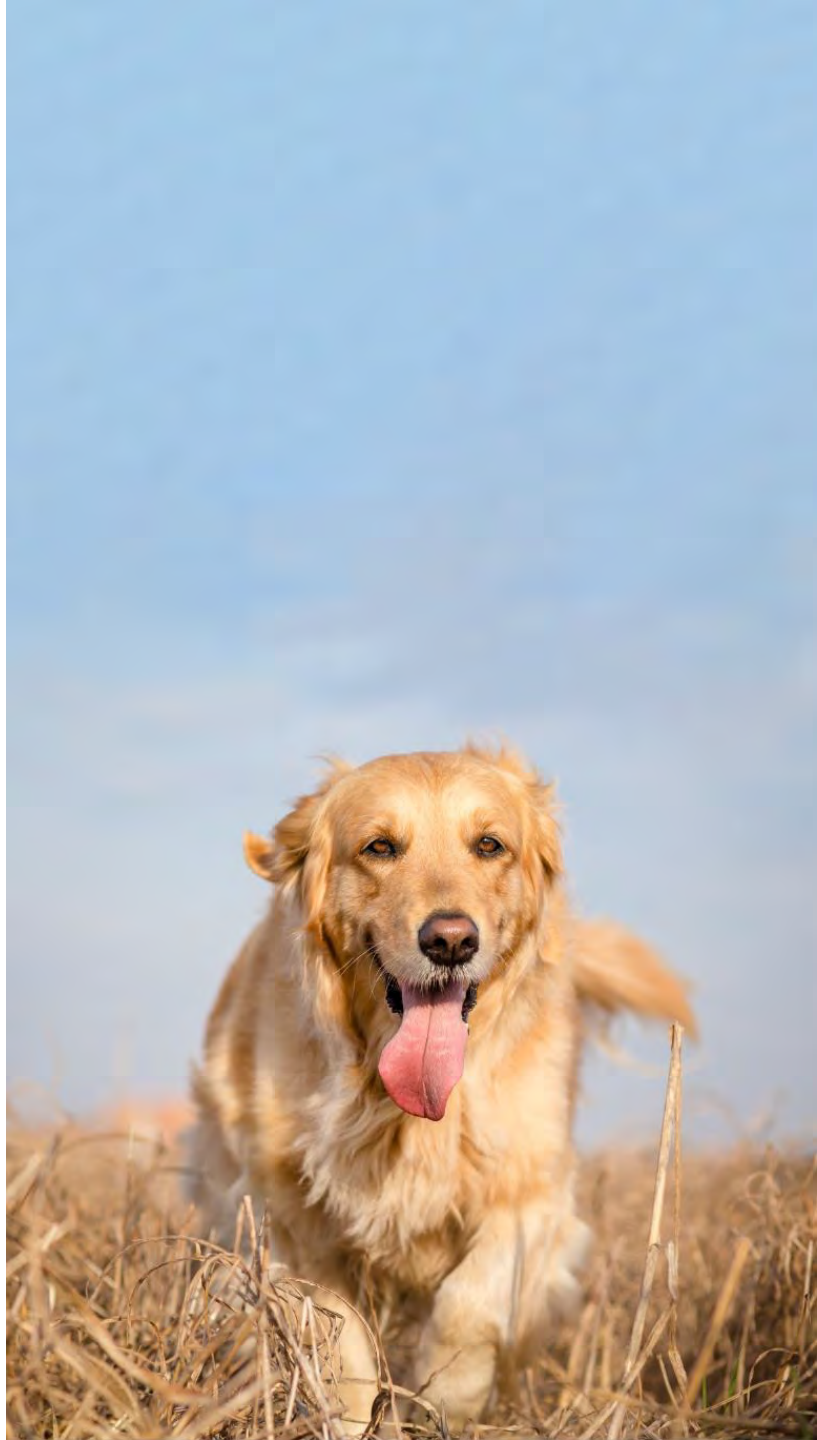
1. Bauden M, Pamart D, Ansari D, Herzog M, Eccleston M, Micallef J, Andersson B, Andersson R. Circulating nucleosomes as epigenetic biomarkers in pancreatic cancer. Clin Epigenetics. 2015 Oct 7;7:106. doi: 10.1186/s13148-015-0139-4. PMID: 26451166; PMCID: PMC4597435.
2. Rasmussen L, Christensen IJ, Herzog M, Micallef J, Nielsen HJ; Danish Collaborative Group on Early Detection of Colorectal Cancer. Circulating cell-free nucleosomes as biomarkers for early detection of colorectal cancer. Oncotarget. 2017 Oct 20;9(12):10247-10258. doi: 10.18632/oncotarget.21908. PMID: 29535803; PMCID: PMC5828191.
3. https://www.prnewswire.com/news-releases/volitions-nuq-vet-assay-detects-two-common-canine-cancers-301049317.html

PetDx

3 M's for Veterinary engagement: What premium spread can "Cancer" command?

SCREEN	OncoK9	ELISA Technology
	Medicine-Centric (SOA genomic diagnostics)	**Panel-Centric**
Medical Value	Assessment of probability of cancer, followed by signal confirmation	Assessment of cancer "risk" within a panel configuration + elevated false positives. Impact from prep
Managerial Operability	Testing two pricing models. Full answer in ten days. Easy workflow	Low-friction pricing. Read may trigger **controversy** with clients. Full answers takes client revisits
Monetary Return	Higher ASP, faster cancer workup potential	Revisit and retesting costs before workups. Vet credibility

- ✓ Can draw from unfasted dogs. No impact of fasting on cfDNA concentration, unlike nucleosome concentration
- ✓ Stable for 7 days at room temp; no need to centrifuge within 1hr or ship overnight on ice



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3. https://www.prnewswire.com/news-releases/volitions-nuq-vet-assay-detects-two-common-canine-cancers-301049317.html

PetDx Bridges to Vet the Latest Proven Human Diagnostic Innovation

NGS-based Liquid Biopsy Universe (Human Medicine) (N > 50)

GRAIL, Agilent, EXACT SCIENCES, freenome, illumina, TEMPUS, Thrive, Exai Bio, NEO GENOMICS, sysmex, GUARDANT, CARIS, Adaptive, CHRONIX, LexentBio, C2i Genomics, Personalis, Biocept, epic sciences, 燃石医学, FOUNDATION MEDICINE, natera, INVITAE, biodesix, BillionToOne, Roche, QIAGEN, Helio, FORESIGHT DIAGNOSTICS, DELFI

➤

Veterinary Medicine (N = 1)

PetDx

HUMAN MEDICINE MILESTONES

Vaccines, X-ray, Antibiotics, Ultrasound, CT Scan, MRI, PCR, Liquid Biopsy

1800 1825 1850 1875 1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000 2010 2020

Vaccines, X-ray, Antibiotics, Ultrasound, MRI, CT Scan, PCR, OncoK9 by PetDx

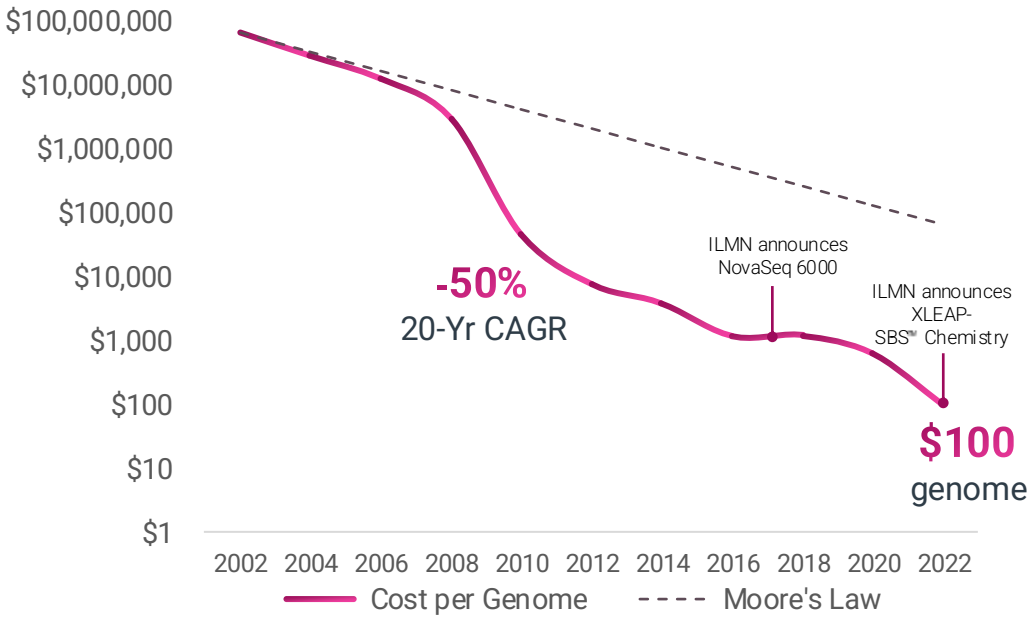
VETERINARY MEDICINE MILESTONES

Sources: Market data

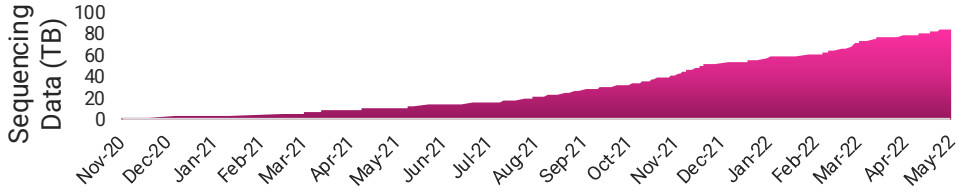
Recent NGS tailwinds unlock profitability and future applications

20-Yr Cost Evolution of NGS (Cost per Genome)

(vs. Moore's Law)



PetDx Genomics Database Growth (>100 TB)



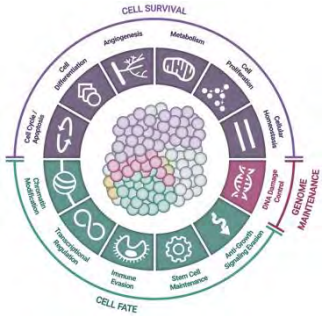
Sources: Market data. https://www.genome.gov/about-genomics/fact-sheets/DNA-Sequencing-Costs-Data

PetDx

Biopharma Applications

Veterinary

Animal



Translational

Human

- New cancer therapeutics
- Patient pipeline
- Targeted treatment and monitoring

- New Human Cancer Tx Development
- Faster & more effective preclinical de-risking of human drug development

Other Applications

- Feline
- NGS/Multi-Omics
- Germline
- Infectious Disease

PetDx®